MIL-D-37079
14 February 1974

MILITARY SPECIFICATION

DEMINERALIZER, WATER, ION EXCHANGE, 115 volt, 60 Hz, AC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of ion exchange water demineralizer that is capable of producing 5 gallons of demineralized water an hour by gravity flow (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-M-7298	Manual, Technical: Commercial Equipment.
MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C73-11 Dimensions of Plugs and Receptacles--General Purpose, 125 volts, 15 Amperes, 2-Pole, 3-Wire, Grounding Type.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Material. Material shall be as hereinafter indicated, and shall be of the highest quality throughout, and shall be suitable in all respects for the purpose intended. All hardware used shall be corrosion-resistant.

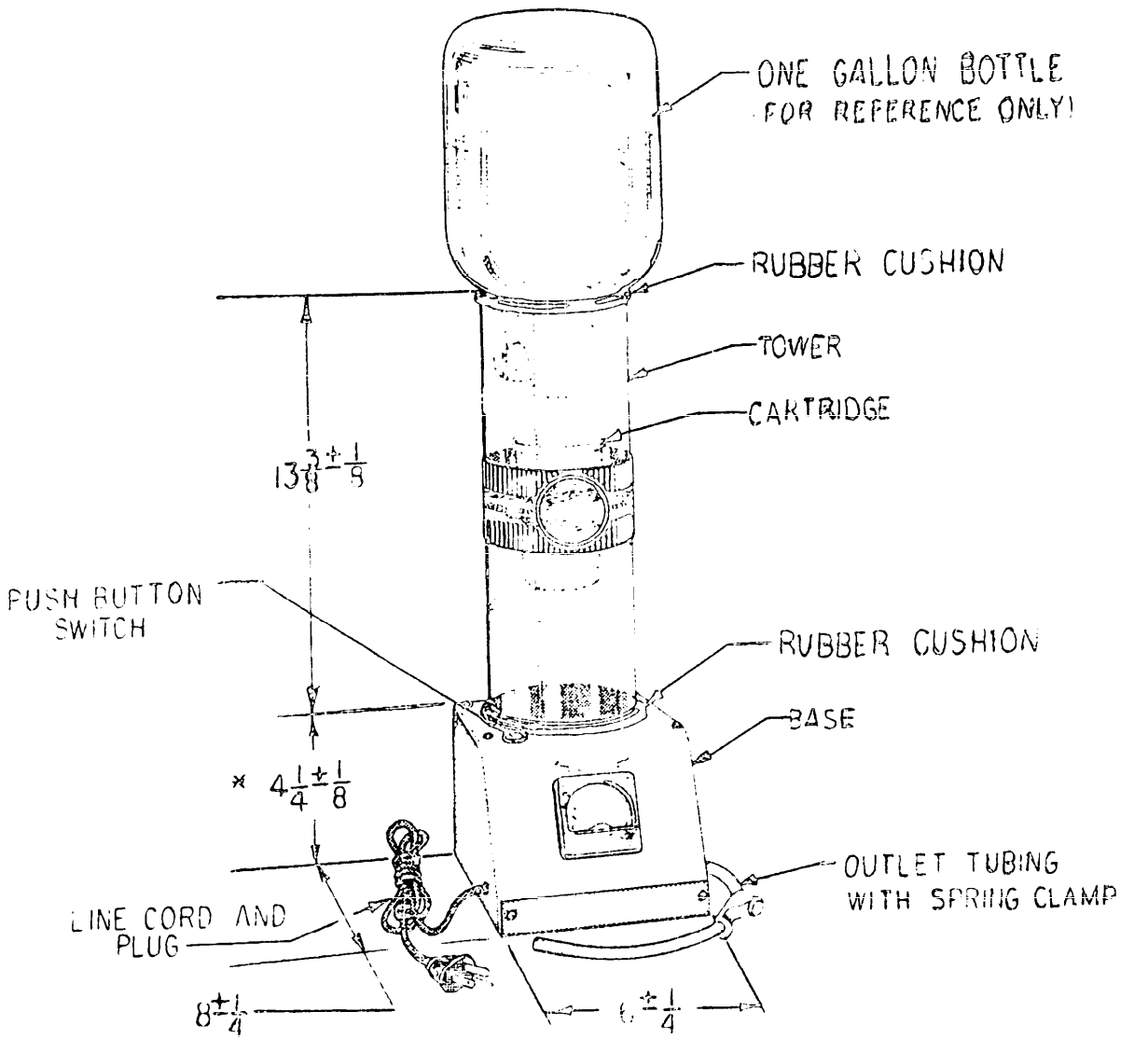
3.2 Design and construction. The demineralizer shall conform to Figure 1 for style and design. Alternate configurations shall be acceptable when not limited by dimensions or specific requirements. The demineralizer shall consist of the following:

a. A shatterproof, transparent plastic tower with two rubber cushions.
b. A base housing.
c. A quality check meter, isolation transformer, and 3-wire electrical cord.
d. Outlet tubing with a spring clamp.
e. Six ion exchange resin cartridge accessories.

The base shall support the lower part of the tower, and shall also house the electrical circuitry for the quality check meter. The weight of the assembled demineralizer shall not exceed 5 pounds.

3.2.1 Tower. The tower shall be fabricated from acrylic plastic, and shall conform to Figure 2 for design and dimensions. The tower shall be a 2-section cylinder that is open at the top and closed at the bottom. The bottom of the cylinder shall have a flat protruding base perpendicular to the vertical axis. The upper and lower sections of the cylinder shall be separated by a shelf with a circular opening to accept, position, and support an inserted ion-exchange resin cartridge (see 3.2.5). The top of the tower shall be provided with a rubber cushion so that a 1-gallon bottle (not required to be supplied with the demineralizer) may sit on top of the tower as indicated in Figure 1,

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* HEIGHT INCLUDES BOTTOM MOUNTED FEET

ALL DIMENSIONS ARE IN INCHES

FIGURE 1

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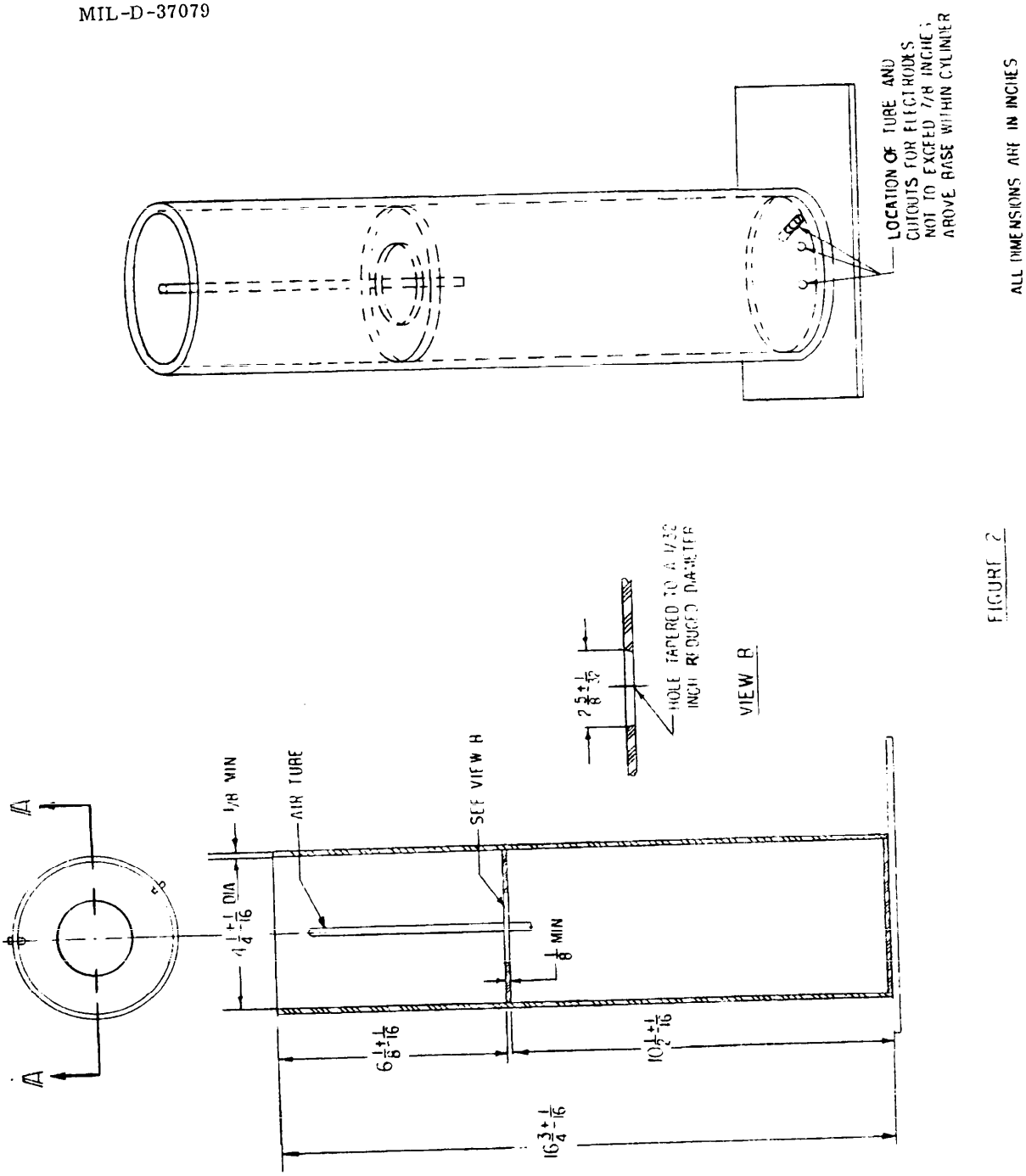


FIGURE 2

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thereby increasing capacity of the demineralizer to 2 gallons. The lower portions of the cylinder shall have cutouts for mounting electrodes. An acrylic plastic air tube shall be supplied which shall extend from a cutout in the wall of the upper section of the cylinder into the lower section of the cylinder through a cutout in the shelf. The shelf, base, outlet tube and air tube shall be firmly attached to the cylinder without any tendency to break off. Four rubber or plastic feet shall be attached to the underside of the protruding section of the base and shall be positioned so that the cylinder is free of rocking and wobbling when resting on a flat level surface.

3.2.2 Base. The base shall be fabricated of 1/32-inch (in.) thick steel sheet. The top of the base shall have a cutout through which the inserted tower can protrude. The cutout shall be provided with a rubber cushion to hold the sides of the tower in a rigid position. The bottom of the base shall be flat so that it can support the bottom of the inserted tower in an upright position. The base shall have additional cutouts for the quality check meter, push-button switch, cord, and outlet tubing (see 3.2.4). The surface with the cutout for the quality check meter shall be inclined at an angle. The steel sheet shall have an enamel finish. The manufacturer's normal commercial color for the finish shall be acceptable. Rubber or plastic feet shall be attached to each corner of the bottom of the base. The base shall be capable of being disassembled to allow access to and removal of the internal components.

3.2.3 Quality check meter circuitry. A direct reading meter shall be used to check the quality of demineralized water. The meter shall operate on 115 ± 5 volt, 60 Hz, AC. An isolation transformer shall be mounted within the base housing. A push-button type switch shall be supplied to activate the meter when readings are required.

3.2.3.1 Quality check meter. The quality check meter shall be mounted on the front side of the base. The meter scale shall be calibrated in both parts per million (ppm.) of sodium chloride and the equivalent resistance in megohms. The scale shall include the following ranges:

a. 0 to 5 ppm. (Infinity to 0.1 megohms)
b. 5 to 9 ppm. (0.1 to 0.05 megohms)
c. 9 to 50 ppm. (0.05 to 0.01 megohms)

When tested in accordance with 4.4.1, the meter scale deflection shall be 5 ± 1/2 ppm.

3.2.3.2 Line cord. A 3-conductor cord shall be supplied with a non-detachable connector plug. The connector plug shall be a 3-blade male plug described in ANSI C73-11. A strain relief shall be provided where the cord enters the base. The overall exposed length of cord and plug shall be not less than 6 feet.

3.2.4 Outlet tubing. A plastic tube shall be supplied and shall be connected to a discharge tube that protrudes from the lower section of the tower. The other end of the tube shall be exposed through a cutout in the base, and shall be supplied with a

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spring clamp to control the flow of water out of the lower section of the tower. The overall exposed length of outlet tubing shall be not less than 9-1/2 in.

3.2.5 Ion exchange resin cartridges. Six cartridges consisting of a mixture of cation and anion exchange resins shall be supplied with each demineralizer (see 6.2). The cartridge shall be constructed of plastic and shall be shaped like a truncated cone. The body of the cartridge shall be filled with resins, and the top and bottom shall have holes to permit water flow (see 3.2.6). A suitable porous filter cloth(s) shall be included in the cartridge to prevent loss of resins but permit the flow of water. The dimensions of the cartridge shall be such that, when inserted in the shelf opening of the demineralizer, it effects a watertight seal around the shelf cutout. The total capacity of each cartridge shall be 180 grains of sodium chloride. The cartridge shall be designed so that, initially and until 144 grains of sodium chloride have been introduced, the cartridge is capable of producing water with an ionic content which does not exceed 5 ppm. of sodium chloride.

3.2.6 Performance. The shelf opening in the tower shall be capable of holding in place a cartridge that is inserted through the top. The inserted cartridge, when secured in the shelf opening, shall effect a watertight seal around the shelf cutout. The demineralizer shall operate by pouring water not exceeding 120° Fahrenheit (°F.) (48.9° Centigrade) through the top of the tower. As the water fills up in the upper section to a level above that of the top of the inserted cartridge, the water shall pass through the cartridge into the lower chamber of the tower. The water that passes through shall be deionized by the ion exchange resins that are contained in the cartridge. The water in the lower chamber shall be removable through the outlet tubing by gravity flow. The spring clamp shall be used to shut off flow of water out of the outlet tubing and allow flow when it is opened. When tested in accordance with 4.4.2, the water shall pass through the cartridge to the outlet tubing, without any evidence of leakage from other parts of the tower. When tested in accordance with 4.4.3, the water passed through the filter shall contain less than 1 ppm. of dissolved salt after 1-1/2 liters accumulation, and shall not exceed 5 ppm. during the entire accumulation.

3.3 Service data. The contractor shall furnish with each unit, two copies each, of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those

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previously accepted, the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the contracting officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

3.4 Workmanship. Workmanship shall be first class throughout. The water demineralizer shall be free from defects which detract from its appearance or may impair its serviceability. The outlet tubing shall not be cut or damaged when the spring clamp has closed off the path. The cartridge shall be free from deformation, tears, cuts, cracks or other defects which detract from its appearance or may impair its serviceability. The water demineralizer shall be free of wobble and rocking when placed on a flat level surface.

3.5 Identification marking. Each water demineralizer shall be permanently and legibly marked with the name or registered trademark of the manufacturer, model number, electrical characteristics (e.g., voltage and frequency), contract number, and "U.S."

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon Government's request, at any time, from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I. The unit of product for inspection purposes shall be the water demineralizer and the individual cartridges.

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Table I For examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II.

Table II For tests

Characteristics	Requirement	Test procedure	Inspection level
Meter deflection	3.2.3.1	4.4.1	100%
Delivery of water	3.2.6	4.4.2	*
Filtration of water	3.2.6	4.4.3	S-2 **

* Three units from each lot of water demineralizers shall be tested. Failure of test unit to pass test shall result in rejection of the lot.

** The cartridges from each lot of ion exchange resin cartridges shall be tested using water demineralizers from each lot being tested in accordance with 4.4.2. Failure of the cartridge or water demineralizer to pass test shall result in rejection of the lot.

4.3 Examination. The water demineralizer and cartridge shall be examined to determine compliance with requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects. Examination shall not be restricted to the classified possible defects listed in Table III.

Table III Classification of defects

Categories	Defect
Major	
Water demineralizer	
101	Unit not complete with all component parts.
102	Base not free of wobble and rocking when placed on a flat surface.
103	Base not free of sharp edges, burrs, cracks, tears or corrosion.
104	Rubber cushions not adequately secured to the plastic tower.
105	Tower not capable of being positioned in an upright position when base is placed on a flat surface.

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Classification of defects (cont'd)

Categories	Defect
Major	
Water demineralizer	
106	Line cord not adequately secured within base.
107	Line cord or rubber cushions not free of cuts, tears or abrasion.
108	Blades of line cord plug not properly aligned.
109	Meter not properly secured to base.
110	Dial graduation on meter not complete, clear or permanent.
111	Spring clamp not capable of being secured or released on outlet tubing.
112	Outlet tubing not free from cracks, tears, cuts or obstructions throughout its length.
Minor	
201	Base not free of dents or deformation.
202	Identification marking not complete, legible or permanent.
203	Unit not free of dirt, grease or foreign matter.
Major	
Cartridge	
101	Cartridge not free of deformation, tears, cuts or cracks.
102	Cartridge cannot be secured to or removed from holder in water demineralizer.
103	Cartridge not free of obstructions to water flow.
104	Components of cartridge not properly secured to body.
Minor	
201	Identification marking not complete, legible or permanent.
202	Outside of cartridge not free of dirt, grease or foreign matter.

4.3.2 Dimensional examination. The demineralizer shall be examined for defects in dimensions. Any dimension not within specified limits shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Meter deflection. The quality check meter line plug shall be connected to a 115 ± 2 volt source. With 80,000 ohms of resistance across the meter electrodes, the meter shall be activated and the meter deflection shall be recorded.

4.4.2 Delivery of water. The demineralizer shall rest on a flat level surface. The spring clamp on the outlet tubing shall be set to the closed position. A cartridge shall be inserted and secured in the water demineralizer. One gallon of water shall be poured into the top of the demineralizer. The demineralizer shall be checked for proper flow through the cartridge to the lower chamber and outlet tubing.

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4.4.3 Filtration of water. The demineralizer shall rest on a flat level surface. The spring clamp on the outlet tubing shall be set to the closed position. A cartridge shall be inserted and secured in the demineralizer. The line cord shall be connected to a 115 ± 2 volt source. Twenty four liters of test solution composed of 3 grams of sodium chloride (NaCl) in 9 liters of water (H_2O) shall be prepared. The test solution shall be slowly passed through the filter cartridge. After 1-1/2 liters of solution has been poured, meter readings (NaCl content in ppm.) shall be recorded for each liter of solution that passes through the filter.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One demineralizer consisting of a plastic tower, base and 6 cartridges, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV Packaging quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	6 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be packaged in a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. The box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents. Closure shall conform to method I as specified in PPP-B-636.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Fiberboard boxes shall conform to special requirements specified in Table II of PPP-B-636.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1, or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II as specified in PPP-B-636. In addition, fiberboard boxes shall conform to the special requirements specified in Table I of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

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5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 The cartridges to be supplied with the demineralizer are intended to conform to the latest specification for NSN 4610-00-921-5622, CARTRIDGE, WATER DEMINERALIZER.

6.3 This specification covers the following item as listed in the Federal Supply Catalog:

National Stock No.	Item Identification
4610-00-782-5867	DEMINERALIZER, WATER, ION EXCHANGE, 115 volt, 60 Hz, AC

6.4 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the type normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 4610-0079

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-D-3709

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
a. GIVE PARAGRAPH NUMBER AND WORDING

b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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